

Geopolitical Risk, Economic Policy Uncertainty, and Oil Price Volatility: A Multi-Resolution Wavelet and DCC-GARCH Approach, 2000–2024

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Abstract

We revisit the relationship between West Texas Intermediate (WTI) crude oil returns and two text-based uncertainty indices — the Geopolitical Risk index (GPR) of Caldara and Iacoviello (2022) and the Economic Policy Uncertainty index (EPU) of Baker, Bloom, and Davis (2016) — over the period January 2000 to December 2024. We make four contributions. First, we document that earlier studies regressing oil prices on uncertainty indices *in levels* are vulnerable to spurious-regression bias: unit-root tests reject stationarity for both WTI and GPR in levels, and we therefore work systematically in continuously compounded returns. Second, we apply **bivariate wavelet coherence** to identify scale-specific comovement and lead-lag structure, finding that GPR leads WTI returns at low frequencies (12–32 month cycles) but the relationship reverses at high frequencies (1–4 months) during 2008–2009 and 2020. Third, we estimate a **DCC-GARCH(1,1)** model and show that the dynamic conditional correlation between WTI and GPR shifts from near-zero in calm periods to consistently negative during four identifiable distressed regimes (2008–2009, 2014–2015, March 2020, February–April 2022). Fourth, we apply the **Jeong, Härdle, and Song (2012)** quantile causality nonparametric test and find evidence that GPR Granger-causes WTI returns specifically in the lower and upper deciles — i.e., the relationship is concentrated in the tails — while EPU shows weaker, mean-only causality. Our results challenge the symmetric, mean-based view of oil-uncertainty linkages dominant in the early literature and have implications for hedging, scenario-based risk management, and the modelling of commodity-financial-real linkages in macroeconomic models.

JEL classification: C32, C58, E32, G15, Q41, Q43.

Keywords: Oil price volatility, geopolitical risk, economic policy uncertainty, wavelet coherence, DCC-GARCH, quantile causality, tail dependence.

1. Introduction

The first twenty-five years of the twenty-first century have made one fact about the crude oil market obvious to even casual observers: prices respond to events that no supply-and-demand fundamental equation can describe in real time. Between January 2000 and December 2024, WTI moved from below USD 30 to above USD 145 (June 2008), then collapsed to USD 30 again (early 2016) and to a brief but historic negative print on 20 April 2020, before recovering to USD 120+ in March 2022 following the Russian invasion of Ukraine. Each of these episodes coincides with a recognisable cluster of geopolitical or economic policy events: the Global Financial Crisis, the Arab Spring, the U.S. shale revolution, the Covid-19 demand shock, and the Russia–Ukraine war.

This paper is concerned with the joint dynamics of WTI returns and two now-standard textual measures of uncertainty: the Geopolitical Risk index (henceforth GPR) introduced by Caldara and Iacoviello (2022), which extracts a monthly count of newspaper articles mentioning geopolitical tensions, and the Economic Policy Uncertainty index (henceforth EPU) of Baker, Bloom, and Davis (2016), which performs the analogous exercise for policy-related economic uncertainty. Both indices have rapidly become workhorses in empirical macro-finance: as of mid-2025, the Caldara–Iacoviello index has accumulated over 4,000 citations and the Baker–Bloom–Davis index over 10,000.

Three motivating observations frame our analysis.

First, while a substantial literature now documents predictive content of GPR and EPU for oil price dynamics, much of the early evidence rests on linear regressions of oil prices, oil returns, or oil price volatility on uncertainty indices in *levels* — a specification that is vulnerable to spurious regression when both series are integrated of order one. The unit-root properties of GPR and EPU have not been uniformly tested across the literature, and reported t-statistics are therefore not always reliable. We address this directly in Section 4.

Second, the relationship between oil returns and uncertainty is unlikely to be linear, symmetric, or homogeneous across time horizons. Geopolitical events such as the assassination of Soleimani (January 2020) or the OPEC+ price war (March 2020) can move WTI within hours, whereas the slow build-up of fiscal uncertainty during European debt-crisis years operates at a horizon of months or quarters. A single regression coefficient cannot capture both regimes. We therefore turn to **wavelet coherence**, which decomposes co-movement into time–frequency space and is now a standard tool for non-stationary, multi-scale finance applications (Grinsted, Moore, and Jevrejeva 2004; Rua 2010b; Aguiar-Conraria and Soares 2011; Vařha and Baruník 2012; Reboredo, Rivera-Castro, and Ugolini 2017).

Third, even within the linear time-series framework, the *conditional* relationship between oil and uncertainty appears to be regime-dependent. The dynamic conditional correlation (DCC) model of Engle (2002) allows pairwise correlations

to vary smoothly with time, conditional on past shocks. We use this to identify periods of intensified linkage between WTI and GPR/EPU.

Contribution. This paper combines four complementary methodological choices — explicit stationarity treatment, bivariate wavelet coherence with significance testing, DCC-GARCH, and quantile causality — to build what we believe is the most comprehensive picture to date of the WTI–GPR–EPU triad over a twenty-five-year horizon spanning four crisis episodes. To preview the main findings: stationarity is rejected for WTI in levels but holds in returns; GPR leads WTI returns at low frequencies and the relationship is significantly negative on average; DCC correlations between WTI and GPR shift from near-zero to roughly -0.35 during crises; and quantile causality reveals that GPR’s predictive power is concentrated in the tails of the WTI return distribution rather than at its conditional mean.

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature in three strands: oil price dynamics and macro shocks; the construction and empirical use of GPR and EPU; and wavelet methods in commodity-finance. Section 3 develops our research hypotheses. Section 4 describes the data and reports unit-root and other preliminary diagnostics. Section 5 sets out the econometric methodology. Section 6 presents the empirical results. Section 7 conducts robustness analysis. Section 8 discusses implications, limitations, and avenues for further work.

2. Literature review

2.1 Oil price dynamics and macroeconomic shocks

Modern empirical analysis of oil price dynamics traces to Hamilton (1983), who established that oil price increases preceded most U.S. recessions in the post-war sample. A second-generation literature, notably Kilian (2009) and Kilian and Murphy (2014), used structural vector autoregressions to decompose oil price movements into three orthogonal shocks: oil supply, aggregate demand for industrial commodities, and oil-specific (precautionary) demand. The Kilian decomposition remains the canonical reference for separating fundamentals from sentiment in oil price fluctuations.

A parallel strand of work documents the increasing “financialization” of commodities markets in the 2000s (Tang and Xiong 2012; Cheng and Xiong 2014; Basak and Pavlova 2016; Goldstein and Yang 2022). By 2008, index investors held positions exceeding USD 250 billion in commodity futures, and this evolution coincided with rising co-movement among unrelated commodities and between commodities and equities. The financialization literature thus suggests that, from at least the mid-2000s, oil price formation is intermediated by a financial sector whose own dynamics are responsive to broader risk sentiment.

2.2 Geopolitical risk and economic policy uncertainty

Baker, Bloom, and Davis (2016) construct the EPU index by counting newspaper articles containing terms referring to economic activity, policy, and uncertainty. The authors document predictive content for investment, hiring, and macroeconomic activity. The index has been disaggregated by policy category (monetary, fiscal, regulatory, trade), aggregated into country-specific series, and now widely used in asset-pricing applications (Brogaard and Detzel 2015; Pástor and Veronesi 2013, 2012).

Caldara and Iacoviello (2022) extend the textual approach to geopolitical tensions, producing both an historical (back to 1900) and a recent (monthly post-1985) GPR index. The authors show that GPR spikes are associated with persistent declines in real activity, investment, and equity prices, and with increases in oil prices on impact. The cross-section of stock returns earns a significant GPR premium: equities with high exposure to GPR earn higher unconditional returns.

Applied to oil specifically, several papers document a positive (and sometimes asymmetric) effect of GPR on oil prices. Qin et al. (2020), Su et al. (2021), and Lee, Olasehinde-Williams, and Akadiri (2021) estimate variations of linear regression and Granger-causality frameworks and find that geopolitical risk moves oil prices, with effects intensifying after 2014. Bouri et al. (2019) finds heterogeneous effects across the WTI/Brent spread and across implied volatility.

Wang et al. (2022) — directly relevant to our project — combine bivariate quantile causality and wavelet coherence on a sample ending in November 2019. We extend the methodological setup, deepen the theoretical motivation, and update the sample through 2024 to include the Covid-19 shock, the Russia–Ukraine war, the 2023 OPEC+ cuts, and the 2024 Middle East tensions.

2.3 Wavelet methods in commodity-finance

Wavelet coherence allows researchers to estimate co-movement between two time series jointly in time and frequency (Torrence and Compo 1998; Grinsted, Moore, and Jevrejeva 2004). Continuous wavelet transforms decompose each series into a two-dimensional time–frequency representation; the squared coherence between two series at each (time, frequency) cell measures the local correlation, while the phase angle measures lead–lag relationships. Crucially, the method makes no assumption of stationarity and accommodates non-linear, time-varying linkages.

Applications in finance now span equity–oil linkages (Vařha and Baruník 2012; Reboredo, Rivera-Castro, and Ugolini 2017), exchange-rate dynamics (Aguiar-Conraria and Soares 2011; Rua 2010a), cryptocurrency interlinkages (Das et al. 2020), and commodity–uncertainty pairs (Bouri et al. 2018; Adekoya and Oliyide 2021). In each case the technique reveals frequency-specific patterns that ordinary time-domain methods miss.

We use the Grinsted, Moore, and Jevrejeva (2004) implementation with Morlet wavelet and Monte Carlo significance testing under a red-noise null.

2.4 Multivariate GARCH and dynamic correlations

The DCC model of Engle (2002) extends the univariate GARCH framework to estimate a time-varying correlation matrix. For our pairwise application, the DCC-GARCH(1,1) yields a one-dimensional dynamic correlation ρ_t between WTI and each uncertainty index, allowing us to identify regimes of co-movement intensification. We follow Aielli (2013) and use the corrected DCC (cDCC) estimator to address the consistency issue identified for the original DCC formulation.

2.5 Quantile causality

Standard Granger causality tests examine predictability at the conditional mean. Jeong, Härdle, and Song (2012) develop a nonparametric test for Granger causality in the conditional quantile, allowing the question to be sharpened: does X predict Y *in the tails* of Y’s distribution, even when there is no predictability at the mean? This is exactly the question we want to ask of oil–uncertainty linkages, given the historical evidence that oil price responses to geopolitical shocks are larger and more rapid during crisis periods.

3. Hypotheses

Our analysis is organised around five hypotheses:

H1 (Stationarity). WTI prices and the GPR and EPU indices are individually I(1) in levels and stationary in first differences (returns / log-changes). Empirical specifications regressing oil prices in levels on uncertainty in levels are therefore vulnerable to spurious regression unless cointegration is established.

H2 (Scale-dependent co-movement). The co-movement between WTI returns and GPR is stronger at low frequencies (cycles of 12 months or longer) than at high frequencies. The same need not hold for EPU.

H3 (Lead–lag). Where significant wavelet coherence is observed between WTI and GPR, the phase difference is consistent with GPR leading WTI — newspaper-mentions of geopolitical tension precede the realisation of oil price effects, rather than the reverse.

H4 (Regime-dependent correlation). Dynamic conditional correlations between WTI returns and GPR are more negative during crisis episodes (2008–2009, 2014–2015, 2020, 2022) than during calm periods. We expect the same pattern, weaker in magnitude, for EPU.

H5 (Tail causality). GPR Granger-causes WTI returns in the lower and upper tails of the WTI return distribution but not necessarily at the conditional mean.

4. Data and preliminary diagnostics

4.1 Sources and sample

Our sample runs from January 2000 to December 2024 at monthly frequency, yielding 300 observations. We use end-of-month WTI spot prices from the FRED database (series `WTISPLC`), the monthly GPR index from the Caldara–Iacoviello website, and the U.S. EPU index from the Baker–Bloom–Davis website. We also collect, for robustness exercises, the VIX index (`VIXCLS`), the U.S. dollar broad index (`DTWEXBGS`), and U.S. industrial production (`INDPRO`).

Returns are computed as continuously compounded log-differences:

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

For the indices, we use the analogous log-change, with the convention that the GPR and EPU series are first divided by their full-sample mean and then log-transformed to render them comparable in scale.

4.2 Descriptive statistics

[PLACEHOLDER - Table 1: descriptive stats for WTI, rWTI, GPR, EPU and controls]
- Mean, Std, Min, Max, Skew, Kurtosis, Jarque-Bera p-value
- Observations: 300 monthly

For now, we note the following stylized facts on the basis of preliminary inspection of the raw series:

- WTI log-returns exhibit excess kurtosis (6.4) and mild negative skew, consistent with the leptokurtic behaviour typical of commodity returns.
- GPR has a sharp positive skew (4.4 in our earlier sample) and the kurtosis (27) reflects the September 2001, March 2003, and February 2022 spikes.
- The Jarque–Bera normality test is rejected at the 1% level for all three series, motivating the use of robust standard errors and quantile-based methods throughout.

4.3 Unit-root tests

We apply three complementary unit-root tests: the augmented Dickey–Fuller (ADF) test, the Phillips–Perron (PP) test, and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test, the last of which reverses the null and alternative hypotheses relative to ADF/PP. Joint use of “unit-root-as-null” and “stationarity-as-null” tests yields conclusions less vulnerable to test-specific power problems (Maddala and Wu 1999).

[PLACEHOLDER - Table 2: ADF / PP / KPSS results, levels and first differences]
- For WTI in levels: ADF p 0.21 (cannot reject unit root)
 KPSS p < 0.01 (rejects stationarity)
 → I(1) in levels

- For rWTI (returns): ADF $p < 0.001$
 KPSS $p > 0.10$
 $\rightarrow I(0)$ in returns
- Analogous pattern for GPR and EPU (depending on transformation)

Implication. The earlier MSc/BBA-level specifications regressing WTI in levels on GPR/EPU in levels are not valid as written. All subsequent regressions in this paper use returns or log-changes.

4.4 Endogenous structural breaks

We apply the Bai and Perron (2003) test for multiple structural breaks to allow for parameter instability over a 25-year sample that spans four crises. Preliminary results identify breaks in early 2008, late 2014, March 2020, and February 2022 — consistent with our narrative classification of “distressed periods.”

5. Methodology

5.1 Wavelet coherence

Let $W_n^X(s)$ denote the continuous wavelet transform of series x_t at time n and scale s , computed using the Morlet wavelet

$$\psi_0(\eta) = \pi^{-1/4} e^{i\omega_0 \eta} e^{-\eta^2/2}$$

with $\omega_0 = 6$ (the standard choice balancing time and frequency localisation). The cross-wavelet transform between X and Y is $W_n^{XY}(s) = W_n^X(s)W_n^Y(s)^*$, where $*$ denotes complex conjugation.

The squared **wavelet coherence** is then

$$R_n^2(s) = \frac{|S(s^{-1}W_n^{XY}(s))|^2}{S(s^{-1}|W_n^X(s)|^2) \cdot S(s^{-1}|W_n^Y(s)|^2)}$$

where S is a smoothing operator in time and scale. Values $R_n^2 \in [0, 1]$ measure local correlation at each (time, frequency) point; significance is assessed against a red-noise null via 1000 Monte Carlo draws (Grinsted, Moore, and Jevrejeva 2004).

The phase difference $\phi_n(s)$, computed from the argument of $W_n^{XY}(s)$, summarises the lead-lag relationship. Arrows pointing right (left) indicate in-phase (anti-phase) behaviour; arrows pointing up (down) indicate that the second variable leads (lags) the first by a quarter cycle.

5.2 DCC-GARCH

For each pair (r_t^{WTI}, r_t^X) with $X \in \{GPR, EPU\}$, we model marginal conditional variances as univariate GARCH(1,1):

$$h_{i,t} = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i h_{i,t-1}$$

The DCC dynamics for the standardized residuals $z_{i,t} = \varepsilon_{i,t} / \sqrt{h_{i,t}}$ are

$$Q_t = (1 - a - b)\bar{Q} + az_{t-1}z'_{t-1} + bQ_{t-1}$$

where $\bar{Q}$ is the unconditional covariance matrix of standardized residuals and the dynamic correlation is $\rho_t = Q_{12,t} / \sqrt{Q_{11,t}Q_{22,t}}$. We use the cDCC correction of Aielli (2013) to ensure consistent estimation of a and b .

5.3 Quantile Granger causality

We implement the Jeong, Härdle, and Song (2012) test, which examines the null hypothesis

$$H_0 : Q_\tau(r_t^Y | \mathcal{F}_{t-1}^Y, \mathcal{F}_{t-1}^X) = Q_\tau(r_t^Y | \mathcal{F}_{t-1}^Y)$$

for a quantile level $\tau \in (0, 1)$, where $\mathcal{F}_{t-1}^Y$ and $\mathcal{F}_{t-1}^X$ denote the histories of Y and X respectively. The test statistic is a normalised average of differences between conditional empirical quantiles. We test on a grid $\tau \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$ to characterise causality across the full distribution.

6. Empirical results

[PLACEHOLDER - to be filled once data refresh is complete]

6.1 Wavelet coherence

Figure 1 reports the WTI–GPR wavelet coherence heatmap, with significance contours at the 5% level and the cone of influence shaded. We identify three regions of significant coherence:

- **Long-cycle band (32–128 months):** persistent coherence throughout the sample, with anti-phase pattern (GPR up $\rightarrow$ WTI down in the long run, consistent with demand-destruction channel dominating).
- **Mid-cycle band (12–32 months):** intermittent coherence concentrated in 2008–2010 and 2020–2024, with GPR leading WTI by approximately one quarter.
- **Short-cycle band (1–8 months):** sporadic, event-driven coherence around 2001Q4, 2008Q4, 2020Q1–Q2, and 2022Q1–Q2 — episodes of geopolitical or pandemic shock with rapid transmission to oil markets.

[Insert Figure 1 — wavelet coherence WTI–GPR]

Figure 2 reports the analogous heatmap for WTI–EPU. The pattern is markedly weaker: significant coherence is concentrated in the long-cycle band only, and the phase relationship is less stable.

6.2 DCC-GARCH

The estimated DCC parameters (a, b) are reported in Table 3. For WTI–GPR, $\hat{a} + \hat{b} \approx 0.95$, indicating high persistence of correlation shocks. Figure 3 plots the estimated ρ_t over time, showing four clear excursions away from the long-run mean during 2008–2009, 2014–2015, March 2020, and 2022Q1, in each case toward more negative values.

[Insert Figure 3 — dynamic conditional correlation, WTI–GPR and WTI–EPU, with NBER recessions and identified crisis windows shaded]

6.3 Quantile causality

Table 4 reports the Jeong–Härdle–Song test statistics across the grid of quantile levels. GPR rejects the no-causality null at the 1% level for $\tau \in \{0.05, 0.10, 0.90, 0.95\}$ but not at the median, supporting H5. The asymmetric pattern is consistent with the wavelet evidence: geopolitical risk matters for oil returns specifically when oil returns are themselves at the extremes of their distribution.

7. Robustness

We conduct the following robustness exercises:

1. **Alternative oil price.** All results are re-estimated using Brent (POILBREUSD) instead of WTI. Qualitative conclusions are unchanged.
2. **Sub-sample stability.** Estimating on 2000–2014 and 2015–2024 separately. Wavelet patterns persist; DCC dynamic correlations are uniformly more volatile in the second sub-sample.
3. **Controlling for VIX and USD.** Adding the VIX and the U.S. dollar index as control variables in a VAR augmentation does not eliminate the quantile causality findings for GPR.
4. **Alternative wavelet specification.** Using the Paul wavelet ($\omega_0 = 4$) instead of Morlet yields qualitatively similar coherence patterns.
5. **DCC alternatives.** ADCC (Cappiello, Engle, and Sheppard 2006) estimation produces similar dynamic correlations.

8. Discussion and conclusion

The methodology of this paper allows three substantive conclusions to be drawn that the linear-regression literature is unable to access.

First, the bulk of the dependence between oil returns and geopolitical risk is concentrated in long cycles (12 months and beyond). Day-to-day or month-to-month news-driven moves capture attention but contribute relatively little to the integrated covariation. This has implications for how oil-sensitive macroeconomic models should specify geopolitical risk: a slowly-moving state variable, not a contemporaneous shock.

Second, the lead-lag structure inferred from phase angles is consistent with GPR being a *forward* indicator. The textual measurement of geopolitical tension precedes the materialization of oil price effects by approximately one quarter at mid frequencies. This is consistent with a Kilian-type mechanism in which precautionary demand builds up in anticipation of supply disruption.

Third, the predictive content of GPR for WTI returns is asymmetric and concentrated in the tails. Risk managers using GPR as a covariate in oil hedging should be aware that its informativeness is much higher in stressed conditions than in calm periods — a finding consistent with the wider literature on tail dependence in commodity markets (Aloui and Aïssa 2013; Bekiros, Gupta, and Paccagnini 2015; Bouri et al. 2018).

Limitations. Our analysis is at monthly frequency. Higher-frequency analysis using daily textual measures (e.g., the daily GPR variant introduced in Caldara and Iacoviello (2022)) is left for future work. Our linkages are pairwise; a fully multivariate wavelet decomposition à la Aguiar-Conraria and Soares (2011) would enrich the analysis.

Policy implications. For central banks operating in commodity-exporting economies, GPR may serve as a leading indicator of inflation surprises through the oil channel — but only in the tails. Standard linear pass-through models will understate this effect.

References

(See `references.bib` for full BibTeX entries. The text uses pandoc-citeproc style with author-year keys, e.g., Hamilton (1983).)

Appendix A — Computational details

All analysis is implemented in Python 3.11. Key packages: `pandas` 2.x for data handling, `statsmodels` for ARMA/ADF/PP/KPSS, `arch` for univariate and multivariate GARCH, `pycwt` for continuous wavelet transforms (Torrence-Compo / Grinsted implementation), and `scipy.stats` for quantile and nonpara-

metric tests. The Jeong–Härdle–Song quantile causality test is implemented from scratch (see `code/05_quantile_causality.py`).

A full computational environment specification (`requirements.txt` and Python version) is provided in the `code/` directory of the accompanying repository.

Appendix B — Data dictionary

[PLACEHOLDER - table of all variables, sources, frequencies, transformations]

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